Joint Navigation and Online Aeromagnetic Compensation by Incremental Factor-Graph Smoothing

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***Abstract*— Airborne magnetic-anomaly navigation bounds inertial drift in GNSS-denied flight by matching a scalar magnetometer against an anomaly map. Its central obstacle is the aircraft's own magnetic field, which must be compensated online when no calibration flight is available, the cold-start regime; on a cabin-mounted magnetometer the uncompensated field exceeds the full range of the map signal itself. We cast navigation and compensation as one maximum-a-posteriori problem on a factor graph, carrying the Tolles–Lawson coefficients as time-varying variables alongside the inertial error state, and solve it with a real-time-capable incremental fixed-lag smoother built on iSAM2, at a $300\,\mathrm{s}$ lag and $1\,\mathrm{Hz}$ states, $7.3\,\mathrm{ms}$ to $9.0\,\mathrm{ms}$ per update. The smoother produces two estimates and we report both throughout: a causal estimate with no look-ahead, which is the like-for-like comparison against recursive filters, and a smoothed estimate revised by the full lag, which is the output of a system permitted to lag real time. On the SGL 2020 flight data, across eight cold-start cases spanning four lines, two flights, two maps and two altitude regimes under one untuned configuration, the causal estimate alone beats an identically modelled online-TL EKF and a neural-network-augmented EKF on six of eight cases, and the smoothed estimate is the most accurate on all eight, at $10.5\,\mathrm{m}$ to $28.4\,\mathrm{m}$ where free-inertial drift is hundreds of metres. A lag sweep locates the design knee near $300\,\mathrm{s}$ and shows the causal estimate is indifferent to the lag, so look-ahead buys smoothing accuracy only. A projected-gradient separability metric, derived from a condition that does not require the redundant compensation coefficients to be identified, falls by one and a half to two orders of magnitude across the $30\,\mathrm{s}$ to $300\,\mathrm{s}$ gain regime and identifies the weakest flight line.**

***Index Terms*— Magnetic-anomaly navigation, aeromagnetic compensation, Tolles–Lawson, factor-graph optimization, fixed-lag smoothing, GNSS-denied navigation.**

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Nomenclature

$\mathbf{x}_t$	INS error state at epoch t (position, velocity, attitude, baro-inertial pair, IMU biases, FOGM), $\mathbf{x}_t \in \mathbb{R}^{18}$.
$\boldsymbol{\beta}_t$	Online Tolles–Lawson compensation coefficients.
$\boldsymbol{\chi}_t$	Joint state $(\mathbf{x}_t, \boldsymbol{\beta}_t)$.
$\boldsymbol{\Phi}_t, \mathbf{Q}_t$	Pinson state-transition matrix and process covariance.
z_t, η_t	Scalar magnetometer measurement and its noise, variance R.
$h(\cdot)$	Anomaly-map interpolation (with IGRF core) as a function of position.
$\mathbf{g}_t$	Map gradient $\partial h/\partial \mathbf{p}$ at epoch t.
$\mathbf{A}_t$	Tolles–Lawson basis row from the three-axis fluxgate.
c_t	Aircraft magnetic interference, $c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t$.
L_w, L_o	Sliding-window length and overlap (look-ahead).
ρ, δ	Robust kernel and its Huber threshold.
$\mathbf{G}, \boldsymbol{\Psi}$	Stacked map-gradient and TL-regressor matrices over a window.
$\mathcal{N}(\cdot,\cdot)$	Gaussian density.
$\|\cdot\|_{\mathbf{M}}$	Mahalanobis norm, $\|\mathbf{v}\|^2_{\mathbf{M}} = \mathbf{v}^\top \mathbf{M} \mathbf{v}$.

Boldface lowercase denotes vectors and boldface uppercase matrices.

I. Introduction

INERTIAL navigation systems (INS) are the backbone of airborne navigation, but their position error grows without bound as accelerometer and gyroscope errors are integrated over time, so an INS must be aided by an absolute position reference [1], [2]. The satellite-based reference of choice, GNSS, is unavailable or untrustworthy in the contested and signal-degraded environments that increasingly define aerospace and defense operations. This has renewed interest in passive, self-contained references that draw on natural geophysical signatures of the Earth. Among these, the crustal magnetic-anomaly field is attractive: it is globally available, temporally stable over the mission horizon, and observable with a lightweight scalar magnetometer [3], [4]. It is also difficult to spoof from a stand-off distance. Denial of a satellite signal exploits the far-field radio link, whose power falls off only as the inverse square of range, so a modest transmitter jams or spoofs a receiver from far away; a magnetic anomaly, by contrast, is a near-field source whose dipole field decays as the inverse cube of distance, so counterfeiting a credible anomaly at a moving aircraft would demand an impractically large or close field source [5]. The same map-matching idea underlies terrain-referenced and gravity-referenced navigation, where an onboard measurement is matched against a stored geophysical map to bound inertial drift [6], [7], [8]; magnetic-anomaly navigation is distinguished by the ubiquity and fine spatial structure of the crustal field and by the maturity of airborne magnetic surveying. These properties make it a candidate absolute reference for long-endurance aircraft, uncrewed platforms, and munitions operating where GNSS cannot be trusted, provided the aircraft's own magnetic signature can be removed in flight.

Magnetic-anomaly navigation (MagNav) estimates aircraft position by matching the measured scalar field against a surveyed anomaly map, yielding an absolute fix that bounds inertial drift [4]. The idea has matured from feasibility study to flight demonstration: recursive Bayesian map matching, from the extended Kalman filter to grid and particle estimators, was characterized on flight data [3], [4] and flown on an F-16 [9], and recent multi-hour fixed-wing trials over thousands of kilometres report metre-to-decametre accuracy with quantum magnetometers [5]. As the estimators have matured, attention has moved to the inputs they rely on, and a consensus is forming that the pacing requirement for operational deployment is a sufficiently accurate, uncertainty-aware anomaly map [10]. Two obstacles, in fact, separate these demonstrations from routine use, and they are distinct: the reference map on the ground and the aircraft's own field in the air. The map is a survey-and-data problem, pursued elsewhere [10]; this paper addresses the second.

The platform's own permanent and induced magnetization and its eddy-current fields perturb the scalar measurement by hundreds of nanotesla, comparable to or larger than the anomaly signal being matched, so the aircraft field must be compensated before map matching can recover position. This aircraft-field problem is intrinsic to MagNav and independent of any particular airframe; it is most acute for magnetometers mounted inside the cabin, close to avionics and current-carrying structure, rather than on a wingtip stinger boom that holds the sensor away from the platform field.

Aeromagnetic compensation is classically handled by the Tolles–Lawson (TL) model [11], [12], [13], a regression of the interference onto sixteen attitude-dependent terms (permanent, induced, and eddy-current) formed from a three-axis fluxgate magnetometer. It faces two difficulties that are easy to conflate but are in fact orthogonal. The first is temporal: the coefficients are conventionally identified on a dedicated high-altitude calibration flight of prescribed maneuvers, yet many operational scenarios admit none, and the coefficients drift with temperature and configuration. In this cold-start regime they must be estimated online and jointly with navigation, from the same measurements that fix position, a coupling that makes both estimation and its analysis substantially harder than the calibrated case. The second is representational: however well its coefficients are fit, the linear TL basis cannot span interference from platform currents and other non-rigid, non-attitude sources, which contribute a large fraction of the field on cabin-mounted sensors [14], [15]. One difficulty is about identifying a known model class in time; the other is about that class being too small.

A public benchmark has made both concrete: the Signal Enhancement for Magnetic Navigation (SGL 2020) challenge dataset [14], with flight-grade INS, wingtip-stinger and cabin-mounted magnetometers, and high-resolution anomaly maps. Most work on it attacks the two difficulties offline: the TL coefficients by a batch calibration flight, and the representational gap by learned models fit offline on logged data, neural-network heading-error and denoising compensators [16], [15], physics-informed liquid-time-constant networks [17], and model-stability studies of the regression in gradiometry [18]. Offline learning, however, presupposes exactly what the operational scenario lacks: a corpus of logged flights from the same aircraft, a reference-grade compensated sensor flying alongside to supply training targets, the computation to train on that corpus, and a willingness to retrain whenever the platform configuration or its magnetic state drifts. A compensator fit this way is specific to one airframe in one equipment state, and its accuracy in flight is bounded by how well the training corpus transfers to the day's conditions. We therefore treat the offline family as context rather than competition: every estimator compared in this paper starts, as ours does, from no prior flight data. The exception is the online NN-in-EKF line of work originated by Gnadt [15], which carries the TL coefficients and the weights of a residual network as filter states learned in flight, and which has recently been extended with an error-state formulation and a natural-gradient interpretation [19]; this lineage holds, to our knowledge, the strongest reported cold-start results on the benchmark. This paper takes on the temporal difficulty fully: online, joint estimation of a time-varying compensation. The representational gap it does not model term-by-term: instead of a learned remainder, the estimator absorbs what the TL basis misses structurally, through the time-varying coefficients, a colored disturbance state, and a robust kernel, and Section V verifies against the recorded platform currents how much of the remainder this accounts for.

Recursive Bayesian filtering is the standard machinery for map-matching navigation, from the linear Kalman filter [20] to nonlinear variants. Geophysical map matching has a long lineage: terrain-contour matching for cruise missiles [6] and nonlinear Kalman filtering for terrain-aided navigation [7] established the paradigm, and grid- and particle-based Bayesian estimators later became standard where the measurement-to-position map is strongly nonlinear or multimodal [21], [22], [23], [8]. For MagNav specifically, Canciani and Raquet combined a calibrated scalar magnetometer with an extended Kalman filter (EKF) and characterized the achievable accuracy on flight data [3], [4], later demonstrating airborne magnetic navigation with online calibration [9]. Beyond the EKF, unscented filtering [24], invariant formulations that respect the state manifold [25], and mixture/particle representations [26] improve consistency under nonlinearity. The invariant line has reached this application: Hager et al. formulate map-matching aiding as an error-state filter on

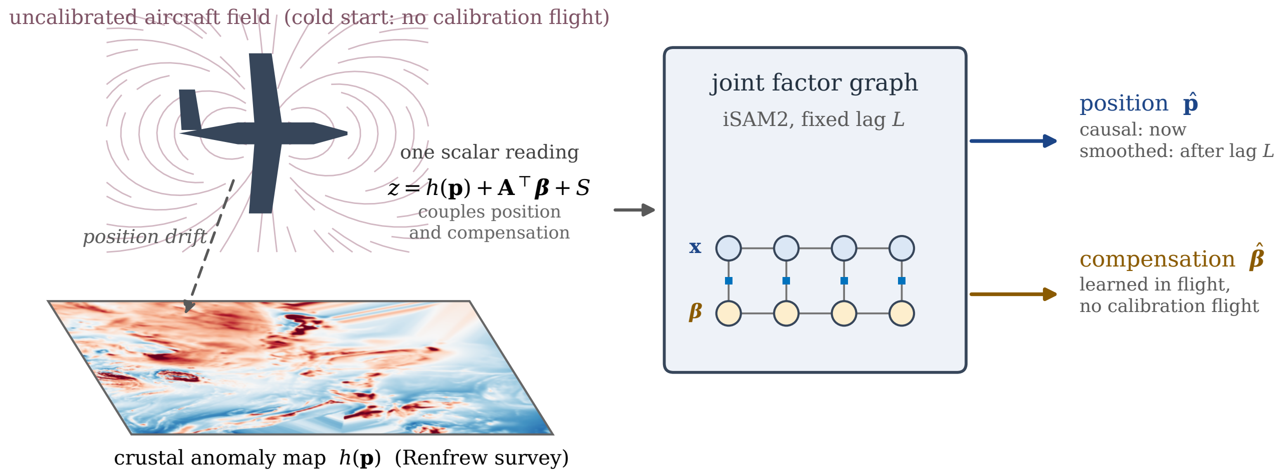

Fig. 1. Concept: one scalar reading couples position and compensation; a joint fixed-lag factor graph estimates both, emitting a causal and a smoothed answer.

the $SE_2(3)$ group of extended poses, correcting the inertial solution in closed loop so that the linearized dynamics stay accurate even for a tactical-grade unit [27]. The group structure improves the propagation and the closed-loop error dynamics, but it does not directly resolve the position–compensation ambiguity studied here, and the map sensitivity remains governed by the local spatial gradient; the study is in simulation and assumes every magnetic source but the anomaly field to be perfectly corrected beforehand, so the aircraft field never enters. Its degradation on low-information maps is consistent with the map-information floor formalized in Section B. However, these estimators are causal: they commit each state online with no access to future measurements, and the position-centric ones carry no compensation state of their own. When a large uncompensated cold-start residual enters, a causal filter has no mechanism to attribute the mismatch to the aircraft field rather than to position, and, unable to revise a state once made, the estimate can diverge.

A second line of work makes the compensation itself adaptive. Classical Tolles–Lawson (TL) identification is a batch regression on a calibration flight [11], [12], [13]; to remove that requirement the TL regression can be embedded in the estimator and its coefficients tracked online as additional states. Nonlinear extensions replace or augment the linear basis with learned components: neural-network compensators date back three decades [16], model-stability and robustness of the regression have been studied in gradiometry [18], and machine-learning-enhanced calibration has recently been applied to the SGL benchmark [15]. Most relevant here, a residual neural network (NN) has been added on top of an online TL model, carried inside an EKF and trained online, to capture interference the linear basis misses, introduced with Gnadt's online models [15] and recently extended in [19]; this reaches good cold-start accuracy on the SGL 2020 cabin magnetometers and holds, to our knowledge, the strongest reported cold-start results on that benchmark. Nevertheless, the network requires a carefully engineered online-training design to remain stable, adds opaque parameters whose behavior is hard to certify, and, being embedded in an EKF, inherits the same causal, commit-online limitation as the filters above.

A third body of work, largely from robotics, replaces filtering with factor-graph optimization: the estimation problem is a graph of variables and probabilistic constraints [28] whose maximum-a-posteriori (MAP) solution is found by sparse nonlinear least squares [29], [30], the same structure underlying bundle adjustment and graph SLAM [31], [32], [33]. Incremental and square-root smoothers (square-root SAM [34], iSAM [35], iSAM2 [36], and exactly-sparse delayed-state filters [37]) make the solution efficient enough for online use, and fixed-lag sliding-window optimization [38], [39] bounds its cost. On-manifold inertial preintegration [40], factor-graph inertial fusion [41], and modern visual–inertial estimators [42] have brought the approach firmly into navigation. Despite these advances, factor-graph estimation in navigation has been developed almost entirely for visual–inertial odometry and simultaneous localization and mapping, where the estimated quantities are poses and landmarks. The factor graph has very recently reached aeromagnetic compensation itself: Lathrop et al. [43] calibrate a magnetometer by fusing vector and scalar magnetometers with inertial measurements in a factor graph, absorbing large permanent moments and a time-varying external field. That work compensates the sensor but does not close the loop to navigation. The joint problem, estimating a time-varying compensation together with aircraft position from map matching, has not, to our knowledge, been posed in a published factor-graph formulation, and the observability of doing so has not been characterized.

Taken together, the three lines leave a specific gap. The cold-start MagNav problem is fundamentally a joint estimation of position and a time-varying compensation model from a single scalar stream, in which the two unknowns are confounded through the very measurement that must resolve them. Causal filters (cluster A) must commit each state before the maneuvers that would disambiguate the compensation have occurred, and can diverge when the initial residual is large; learned online compensation (cluster B) repairs that residual but keeps the causal, commit-online architecture and adds parameters that are difficult to certify; and factor-graph smoothing (cluster C), which can defer commitment and estimate structured nuisance variables jointly, has not been applied to this problem, nor has anyone characterized when the joint problem is even observable. What is missing is an estimator that (i) defers commitment so that late-arriving calibration information corrects early navigation states, (ii) represents the compensation as first-class, time-varying variables rather than a black box, and (iii) comes with an explicit condition for when position and compensation can be separated at all. This paper supplies all three.

In this work we cast cold-start magnetic-anomaly navigation as a factor graph and, within it, carry the online Tolles–Lawson coefficients as time-varying variables, so that navigation and compensation are solved jointly as one maximum-a-posteriori (MAP) problem (Fig. 1). The graph is solved in real time by an incremental fixed-lag smoother built on iSAM2 [36]: states are placed at $1\,\mathrm{Hz}$, variables older than a $300\,\mathrm{s}$ lag are marginalized, and each update touches only the part of the factorization the new measurement disturbs. Every state inside the lag is relinearized and revised as later measurements arrive, so late calibration information corrects early navigation states, the mechanism a causal filter lacks; a causal filter commits each state once and cannot revise it. We are explicit about scope. For a linear–Gaussian model the fixed-lag MAP estimate coincides with classical fixed-lag smoothing [44], so smoothing per se is not a contribution, and the incremental solver is standard machinery used as published [36]: what is new is the formulation that the solver is applied to, and what it makes possible. That the resulting estimator runs comfortably in real time, at $7.3\,\mathrm{ms}$ to $9.0\,\mathrm{ms}$ per state for a $300\,\mathrm{s}$ lag at $1\,\mathrm{Hz}$ states, is reported as a feasibility result rather than claimed as novelty. The paper makes the following contributions:

1) **Joint navigation and compensation on a factor graph.** We carry the online Tolles–Lawson coefficients as time-varying variables of a factor graph so that navigation and compensation are estimated jointly from the same scalar measurements. To the authors' knowledge this is the first factor-graph formulation that jointly estimates time-varying Tolles–Lawson compensation and navigation states at a calibration cold start. Relinearizing the coefficients as the graph is re-solved makes the compensation adapt along the flight, the role a learned online model plays in the competing filter, while the graph keeps every quantity an explicit, physically interpretable state.
2) **A separability condition and a flight-dependent lag metric.** We give a necessary and sufficient local condition for separating horizontal position from the compensation subspace, one that does not require the redundant Tolles–Lawson coefficients to be uniquely identified, and turn it into a computable weakest-direction information metric. Evaluated on the flight lines, it explains where the accuracy gained by lengthening the lag comes from and which line is hardest, and it indicates the transition beyond which the measured error is no longer predominantly separability-limited.
3) **Flight-data validation against reproduced baselines.** On uncompensated cabin magnetometers across eight cold-start cases spanning four lines, two flights, two maps and two altitude regimes, under one untuned configuration, the smoothed estimate is the most accurate of the compared estimators on all eight cases and the causal estimate on six. The baselines are run rather than cited: the online EKF+TL+NN lineage of [15], [19] is reproduced and tuned for the cold start, and the recursive EKF is given the proposed estimator's own compensation prior, yet the smoother beats both with no learned component. Because a fixed-lag smoother produces a causal and a smoothed estimate and comparisons in the literature rarely distinguish them, we report both throughout, so that the comparison against causal filters is like for like and the price of latency is explicit.

The remainder of this paper is organized as follows. Section II states the problem in its classical batch form and formalizes the cold start. Section III sets out factor graph optimization from first principles. Section IV develops the joint system model, its factor graph, and the incremental fixed-lag solution. Section V describes the protocol and baselines and reports the experiments. Section VI concludes.

II. Problem Formulation

This section states what magnetic-anomaly navigation asks for in its classical batch form, shows why the aircraft's own field prevents that form from being answered directly, and states the problem solved in the rest of the paper.

A. Batch map matching

Database-referenced navigation determines position by comparing a measured profile against a stored map. Let $\mathbf{p}_{0:k}$ be the horizontal positions of the aircraft over $k+1$ epochs, $z_{0:k}$ the scalar readings, and $h(\cdot)$ the map lookup returning the field predicted at a position. The classical batch estimate minimizes a profile-matching criterion, in mean absolute or mean squared form,

$$J_{\mathrm{MAD}} = \frac{1}{k}\sum_{i=0}^{k}\big|\, z_i - h(\mathbf{p}_i)\big|, \quad J_{\mathrm{MSD}} = \frac{1}{k}\sum_{i=0}^{k}\big(z_i - h(\mathbf{p}_i)\big)^2, \tag{1}$$

subject to the inertial navigation system supplying the increments between consecutive positions,

$$\mathbf{p}_i - \mathbf{p}_{i-1} = \Delta\mathbf{p}_i^{\mathrm{ins}}, \qquad i = 1, \ldots, k, \tag{2}$$

so that the search is over a rigid profile translated by the single unknown initial state,

$$\hat{\mathbf{p}}_{0:k} = \arg\min_{\mathbf{p}_{0:k}} J. \tag{3}$$

Terrain-referenced navigation is solved in this form routinely [45], [46], and the same form underlies the magnetic case [3], [4].

We state Eqs. (1)–Eq. (3) as the reference formulation of database-referenced navigation rather than as a competing method: airborne magnetic-anomaly navigation is in practice run recursively, with a Kalman filter or a particle filter aiding the inertial solution [4], [15], and those are the baselines compared against in Section V. The batch form is worth writing down because it makes two things explicit that the recursive implementations leave implicit, and both matter here. It shows what the map contributes, namely the agreement between a measured profile and a stored one. And it treats the inertial increment as the hard constraint Eq. (2), which is the assumption the factor graph of Section III relaxes by admitting the increment as one more piece of uncertain information.

What Eq. (1) presumes is that the residual $z_i - h(\mathbf{p}_i)$ is small and unstructured once the position is right. For a scalar magnetometer mounted on an aircraft this is false, and by a wide margin. The measurement carries the aircraft's own magnetic field, generated by permanent magnetization, by currents induced in the airframe, and by eddy currents, all of which depend on the aircraft attitude in the Earth field and therefore change continuously along a line. On the four flight lines used in Section V the map value along the flown track varies with a standard deviation of $123\,\mathrm{nT}$ to $327\,\mathrm{nT}$ and a peak-to-peak swing of $586\,\mathrm{nT}$ to $2377\,\mathrm{nT}$: that variation is the entire position signal. Against it, the uncompensated field at a cabin-mounted magnetometer is $1729\,\mathrm{nT}$ to $1978\,\mathrm{nT}$ root-mean-square over the first $300\,\mathrm{s}$ of flight. The interference is not a perturbation of the signal, it is larger than the signal's full range, and Fig. 2 shows the two side by side over the representative line.

Compensation is therefore not a refinement but a precondition. The classical remedy is a calibration flight: a dedicated sequence of maneuvers at high altitude, where the anomaly field is weak, from which the interference model is fit once and then subtracted for the rest of the mission [11], [12], [13]. That remedy is unavailable in the situation this paper addresses. An aircraft that departs with an uncalibrated installation, or whose interference has changed since its last calibration, has no compensated measurement to feed into Eq. (1). Estimating position

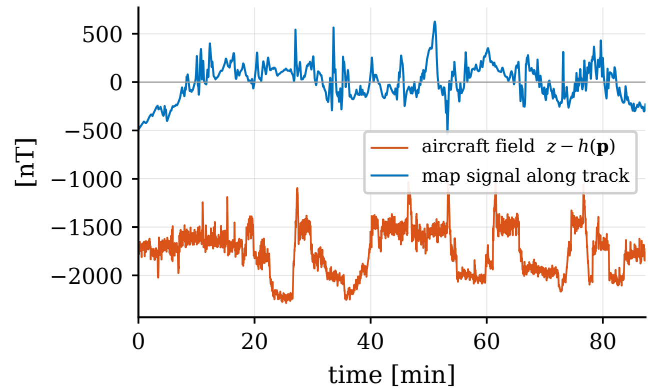

Fig. 2. Line 1007.06 Mag 4: the uncompensated aircraft field exceeds the full range of the map signal along the track.

then requires estimating the compensation at the same time, from the same scalar stream that the map matching consumes, and the two are coupled through that single stream: an error in the compensation is indistinguishable, epoch by epoch, from an error in position. We refer to this situation as a cold start, and it is the setting of this paper.

B. Problem statement

The situation just described fixes what has to be estimated and what may be assumed. The aircraft carries a strapdown inertial navigation system whose error grows without bound, a three-axis fluxgate, and a scalar magnetometer whose reading is corrupted by the aircraft field. A gridded anomaly map of the overflown region is available. No calibration flight has been made, so no compensation model is known at takeoff, and no satellite navigation is available at any point.

Formally, the problem this paper solves is the following: given the scalar magnetometer measurements $z_{0:N}$, the fluxgate regressor rows $\mathbf{A}_{0:N}$, the anomaly map h, the inertial mechanization, and a prior $(\hat{\boldsymbol{\chi}}_0, \mathbf{P}_0)$ that assumes no prior knowledge of the aircraft field, estimate the joint trajectory $\boldsymbol{\chi}_{0:N}$ of inertial navigation errors $\mathbf{x}_k$ and time-varying compensation coefficients $\boldsymbol{\beta}_k$, with a per-epoch cost that does not grow with flight duration, so that the estimate is available in flight.

Three properties of this problem shape the rest of the paper. It is *joint*, because the compensation cannot be measured separately from the position it corrupts. It is *underdetermined at any single epoch*, because one scalar reading cannot separate a position error from a compensation error, so the two are told apart only over a span of epochs. And it is *online*, because an estimator that must see the whole flight before answering is of no use to the aircraft that flew it. The first two properties call for an estimator that carries the compensation as a state and revises past states as later data arrive; the third bounds how far into the past it may keep revising. Sections III and IV construct such an estimator.

III. Factor Graph Optimization

The joint problem of Section B is stated as maximum a posteriori (MAP) inference and solved on a factor graph. This section sets out that machinery from the beginning, since the structure of the graph is what allows the compensation to be carried as a first-class unknown and the estimate to be produced in real time.

A. Variables, factors, and the graph

An estimation problem consists of quantities that are unknown and pieces of information that constrain them. In a factor graph the two are drawn as two kinds of node [47], [29]. A *variable node* holds an unknown, here the state $\boldsymbol{\chi}_k$ at one epoch. A *factor node* holds one piece of information, and an edge joins a factor to every variable that piece of information involves. No edge joins two variables or two factors, so the graph is bipartite, and a factor is a function only of the variables it touches. Fig. 3 shows this pattern for the graph this paper builds in Section IV.

Three kinds of factor appear in any navigation graph, and they exhaust the ones used in this paper. A *prior* factor is unary and states what is believed about a variable before any data: it fixes the otherwise free origin of the estimate. A *process* factor is binary and states how consecutive states are related through the system dynamics and the inertial increments. A *measurement* factor is unary and states what a sensor reading predicts about the state it was taken at. Writing $\mathbf{X} = \boldsymbol{\chi}_{0:N}$ for the collection of variables and $\mathbf{Z}$ for the data, the posterior factors along exactly these edges,

$$p(\mathbf{X} \mid \mathbf{Z}) \;\propto\; p(\boldsymbol{\chi}_0) \prod_{k=1}^{N} p(\boldsymbol{\chi}_k \mid \boldsymbol{\chi}_{k-1}) \prod_{k=0}^{N} p(z_k \mid \boldsymbol{\chi}_k), \quad (4)$$

which is written compactly as a product of factors

$$J(\mathbf{X};\mathbf{Z}) = \prod_i \phi_i(\mathbf{X}_i;\mathbf{Z}), \qquad \hat{\mathbf{X}} = \arg\max_{\mathbf{X}} J(\mathbf{X};\mathbf{Z}), \quad (5)$$

with $\mathbf{X}_i$ the variables adjacent to factor i. The graph is a picture of this factorization, and its sparsity is the problem's sparsity: each factor constrains one or two states, never the whole trajectory at once.

B. From factors to least squares

Each factor is modelled with a Gaussian kernel about its predicted value,

$$\phi_i(\mathbf{X}_i;\mathbf{Z}) \propto \exp\Big(-\tfrac{1}{2} \big\| g_i(\mathbf{X}_i) - \mathbf{z}_i \big\|^2_{\mathbf{P}_i} \Big), \quad (6)$$

where g_i is the prediction associated with the datum $\mathbf{z}_i$, $\mathbf{P}_i$ its covariance, and $\|\mathrm{e}\|_{\mathbf{P}} = \sqrt{\mathbf{e}^\top \mathbf{P}^{-1} \mathbf{e}}$ the Mahalanobis norm. Taking the negative logarithm of Eq. (5) turns the product into a sum and the maximization into a nonlinear least-squares problem,

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{X}} \sum_i \big\| g_i(\mathbf{X}_i) - \mathbf{z}_i \big\|^2_{\mathbf{P}_i}. \quad (7)$$

The Mahalanobis weighting is what makes this sum meaningful. Inertial increments, magnetic readings in nanotesla, and compensation coefficients have different units and differ in magnitude by orders; they become comparable only after each residual is divided by its own uncertainty. The covariance attached to a factor is therefore not bookkeeping, it sets how much authority that piece of information carries against every other; Section E sets the one covariance a cold start is sensitive to.

C. Solving the graph

Because g_i is nonlinear in the states, Eq. (7) is solved iteratively. Linearizing every residual about the current estimate $\mathbf{X}^{(j)}$ and stacking gives

$$r(\mathbf{X}) = \begin{bmatrix} g_1(\mathbf{X}_1) - \mathbf{z}_1 \\ \vdots \\ g_{N_f}(\mathbf{X}_{N_f}) - \mathbf{z}_{N_f} \end{bmatrix}, \qquad \mathcal{G} = \frac{\partial r}{\partial \mathbf{X}}, \quad (8)$$

after which the Levenberg–Marquardt step

$$\Delta\mathbf{X} = -\big(\mathcal{G}^\top \mathcal{G} + \lambda \mathbf{I}\big)^{-1} \mathcal{G}^\top r(\mathbf{X}) \quad (9)$$

is applied and the residuals are relinearized. The damping λ interpolates between Gauss–Newton and gradient descent and makes the iteration robust where the problem is ill conditioned [34], which matters here because the map term is strongly nonlinear.

Two consequences follow, and they are the reason for preferring this formulation over a recursive filter. First, $\mathcal{G}$ inherits the sparsity of the graph: a chain of process factors with unary measurements gives a block-bidiagonal $\mathcal{G}$, so Eq. (9) costs time linear in the number of states rather than cubic. Second, every state is revised at every iteration. A filter commits each state once, at the moment it is reached, and can never revisit it; the graph relinearizes the whole set, so information that arrives late still corrects estimates made early. In the cold start of Section A this is exactly what is needed, because the compensation that explains the early measurements is only identified after the aircraft has manoeuvred.

D. Batch, fixed-lag, and incremental

The formulation says nothing about how much of the flight $\mathbf{X}$ contains, and that choice is the practical design variable. Taking $\mathbf{X}$ to be the whole line gives the batch estimate: it uses every measurement and is the most accurate, but it grows without bound and is available only after landing. Restricting $\mathbf{X}$ to the states inside a bounded interval of the present, and *marginalizing* those that fall outside, gives a fixed-lag smoother whose per-epoch cost does not grow with flight duration [41], [36]. Marginalization is not free: eliminating a variable replaces its factors by a single dense factor on its former neighbours, so what leaves the interval is summarized rather than discarded, and it can no longer be revised.

Rebuilding and re-solving that interval from scratch at every epoch would waste most of the work, since one new measurement changes little of a long chain. Incremental solvers instead update the factorization in place, touching only the part of it that the new data affect [36]. This is what makes a fixed-lag smoother run at sensor rate, and it is the solver this paper uses; Section D states the construction and Section E measures what the length of the interval costs and buys.

IV. Methodology

Section A left the cold start with two unknowns coupled through one scalar stream, and Section III supplied a formulation in which several unknowns of different kinds can be estimated together. This section puts them together: the system model first, then the factor graph it induces, then the incremental fixed-lag solution that produces an estimate in flight.

A. System model

The estimated quantity at epoch k is the joint state

$$\boldsymbol{\chi}_k = \begin{bmatrix} \mathbf{x}_k \\ \boldsymbol{\beta}_k \end{bmatrix} \in \mathbb{R}^{18+m}, \tag{10}$$

stacking the inertial error state $\mathbf{x}_k$ and the compensation coefficients $\boldsymbol{\beta}_k$.

1. Inertial error state

We adopt the standard Pinson error model of a strapdown INS [1], [2], [48]. The error state

$$\mathbf{x} = \left[\, \delta\mathbf{p}^\top \ \delta\mathbf{v}^\top \ \boldsymbol{\psi}^\top \ h_a \ \hat{a} \ \mathbf{b}_a^\top \ \mathbf{b}_g^\top \ S \,\right]^\top \in \mathbb{R}^{18} \tag{11}$$

stacks position error $\delta\mathbf{p}$, velocity error $\delta\mathbf{v}$ and attitude error $\boldsymbol{\psi}$ (three components each), the baro-inertial altitude-aiding pair $(h_a, \hat{a})$ that damps the vertically unstable channel, accelerometer and gyroscope biases $\mathbf{b}_a, \mathbf{b}_g$, and a scalar first-order Gauss–Markov (FOGM) state S that absorbs residual magnetic disturbance and map error. Its discretized dynamics are

$$\mathbf{x}_k = \boldsymbol{\Phi}_{k-1}\mathbf{x}_{k-1} + \mathbf{w}_{k-1}, \qquad \mathbf{w}_{k-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_{k-1}), \tag{12}$$

with $\boldsymbol{\Phi}_k$ the Pinson state-transition matrix (Appendix A) evaluated on the INS-indicated trajectory, and $\mathbf{Q}_k$ the process covariance. Unlike the hard constraint Eq. (2) of the batch form, the inertial increment enters as a factor with a covariance, so the estimator may disagree with the INS in proportion to how much that increment is trusted.

2. Compensation state

Classical aeromagnetic compensation writes the aircraft field as a linear combination of attitude-dependent basis terms [11], [12], [13],

$$c_k = \mathbf{A}_k^\top \boldsymbol{\beta}_k. \tag{13}$$

With direction cosines $\hat{\mathbf{b}}_k = [\, u \ v \ w \,]^\top$ of the Earth field in body axes, measured by a three-axis fluxgate, and total field magnitude B_k, the permanent (3), induced (6) and eddy-current (9) groups are

$$\mathbf{A}_k = \big[\, \underbrace{u, v, w}_{\text{perm.}},\ \underbrace{B_k u^2, \ldots, B_k w^2}_{\text{induced}},\ \underbrace{B_k u\dot{u}, \ldots, B_k w\dot{w}}_{\text{eddy}} \,\big]^\top, \tag{14}$$

optionally augmented by a constant term, giving $m \leq 19$ coefficients. In the calibrated setting $\boldsymbol{\beta}$ is fit once on a calibration flight and frozen. At a cold start it is unknown at takeoff, and the installation state drifts slowly in flight, so we carry it as a variable with a random-walk model

$$\boldsymbol{\beta}_k = \boldsymbol{\beta}_{k-1} + \mathbf{w}_{k-1}^{\beta}, \qquad \mathbf{w}_{k-1}^{\beta} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^{\beta}). \tag{15}$$

This is the same device by which [49], [50] carry slowly varying sensor biases in a terrain-referenced graph. The difference is dimensional: a radar-altimeter bias is one scalar, whereas Eq. (14) is an m-dimensional regression whose columns are driven by aircraft attitude and therefore excited only by manoeuvre.

3. Measurement

The scalar magnetometer reads

$$z_k = h(\mathbf{p}_k) + \mathbf{A}_k^\top \boldsymbol{\beta}_k + S_k + \eta_k, \qquad \eta_k \sim \mathcal{N}(0, R), \tag{16}$$

where $h(\cdot)$ interpolates the crustal anomaly map with the core field restored, evaluated at the true position $\mathbf{p}_k = \bar{\mathbf{p}}_k + \delta\mathbf{p}_k$ formed from the INS-indicated position and the estimated position error. Linearizing the map term about the INS-indicated position,

$$z_k \approx h(\bar{\mathbf{p}}_k) + \mathbf{g}_k^\top \delta\mathbf{p}_k + \mathbf{A}_k^\top \boldsymbol{\beta}_k + S_k + \eta_k, \qquad \mathbf{g}_k = \left.\frac{\partial h}{\partial \mathbf{p}}\right|_{\bar{\mathbf{p}}_k}, \tag{17}$$

which exposes the structure of the problem. The map gradient $\mathbf{g}_k$ is the only channel through which the scalar reading carries horizontal position: where the map is locally flat, $\mathbf{g}_k \approx \mathbf{0}$ and position is momentarily unobservable. Everywhere else, the two terms $\mathbf{g}_k^\top \delta\mathbf{p}_k$ and $\mathbf{A}_k^\top \boldsymbol{\beta}_k$ compete for the same scalar: any position error can be mimicked by a compensation error whose regressor contribution matches it at that epoch. Separating them is possible only over a span of epochs, because $\mathbf{g}_k$ follows the map under the flight path while $\mathbf{A}_k$ follows the aircraft attitude, and the two vary differently. This is the reason a smoother is used rather than a filter, and it is quantified in Section B.

B. Separability of position and compensation

The competition just described can be made exact. Collect, over a window of W epochs, the horizontal position sensitivities (the vertical channel is baro-aided) and the regressor rows of the linearized measurement Eq. (17) into

$$\mathbf{G} = \begin{bmatrix} \mathbf{g}_1^\top \\ \vdots \\ \mathbf{g}_W^\top \end{bmatrix} \in \mathbb{R}^{W\times 2}, \qquad \boldsymbol{\Psi} = \begin{bmatrix} \mathbf{A}_1^\top \\ \vdots \\ \mathbf{A}_W^\top \end{bmatrix} \in \mathbb{R}^{W\times m}, \tag{18}$$

so that a constant position perturbation $\delta\mathbf{p}$ and a constant coefficient perturbation $\delta\boldsymbol{\beta}$ change the residual sequence by $\mathbf{G}\,\delta\mathbf{p}$ and $\boldsymbol{\Psi}\,\delta\boldsymbol{\beta}$ respectively.

Proposition 1 (Position separability):
A position perturbation $\delta\mathbf{p} \neq \mathbf{0}$ is distinguishable from every compensation adjustment over the window if and only if $\mathbf{G}\,\delta\mathbf{p} \notin \operatorname{range}(\boldsymbol{\Psi})$. Consequently every nonzero $\delta\mathbf{p}$ is distinguishable, so that the pair $(\delta\mathbf{p}, \delta\boldsymbol{\beta})$ is locally identifiable up to $\delta\boldsymbol{\beta} \in \ker\boldsymbol{\Psi}$, and in particular horizontal position is locally identifiable, if and only if

$$\operatorname{rank}\big[(\mathbf{I} - \boldsymbol{\Psi}\boldsymbol{\Psi}^{\dagger})\,\mathbf{G}\big] = 2, \tag{19}$$

where $\dagger$ denotes the Moore–Penrose pseudoinverse.

The proof is in Appendix C. The condition deliberately does not ask for identifiability of $\boldsymbol{\beta}$ itself, which would require $\ker\boldsymbol{\Psi} = \{\mathbf{0}\}$, because the 19-term basis does not satisfy it: in continuous time $B_k(u\dot{u}+v\dot{v}+w\dot{w}) = 0$ ties the three diagonal eddy columns. This identity is exact in continuous time and is weakly violated by the finite-differenced fluxgate rates. The induced columns are tied differently: with normalized direction cosines $B_k(u^2+v^2+w^2) = B_k$ holds sample by sample exactly, so their sum equals the total-field column, which is collinear with the constant bias column only up to the variation of B_k, about one percent RMS over a line. Both ties survive on the data, since with every column normalized to unit norm the smallest singular value of a $300\,\mathrm{s}$ window of $\boldsymbol{\Psi}$ is still six orders of magnitude below the largest. Navigation, however, needs the compensation signal $\boldsymbol{\Psi}\boldsymbol{\beta}$, not the coefficients: two coefficient vectors differing by an element of $\ker\boldsymbol{\Psi}$ produce the same correction, so the gauge is harmless to position, and Eq. (19) is the condition that matters. The pseudoinverse makes the condition gauge-invariant by construction: the projector is unchanged by any reparameterization of the coefficients that preserves the compensation column space. Numerically, the metric below is unchanged across four decades of rank tolerance despite the near-null directions. The FOGM disturbance S_k of Eq. (17) competes for the same scalar as well: its window-constant part lies in $\operatorname{range}(\boldsymbol{\Psi})$ through the bias column and is projected out with it, while its variation inside the window is neglected in this analysis.

The condition also reads directly onto the estimator design. At a single epoch, $\mathbf{G}$ has one row and the projected matrix has rank at most one, so Eq. (19) can never hold: a memoryless correction cannot separate the two effects. As the window grows, the map gradient under the flight path and the attitude-driven regressor decorrelate and the projected gradient accumulates rank and strength; the smallest eigenvalue of the Gram matrix of the whitened projected gradient (the squared smallest singular value),

$$\mu(W) = \lambda_{\min}\Big(\bar{\mathbf{G}}^{\top}(\mathbf{I} - \bar{\boldsymbol{\Psi}}\bar{\boldsymbol{\Psi}}^{\dagger})\bar{\mathbf{G}}\Big), \tag{20}$$

with $\bar{\mathbf{G}} = R^{-1/2}\mathbf{G}$ and $\bar{\boldsymbol{\Psi}} = R^{-1/2}\boldsymbol{\Psi}$ the noise-whitened matrices, is a computable, flight-dependent measure of that strength: $\mu(W)^{-1/2}$ is the weakest-direction position standard deviation that a single isolated window of W epochs supports under a static-coefficient reading of the model. It is not a bound on the estimator's error, and its assumptions cut both ways. It is optimistic in holding $\delta\mathbf{p}$ and $\boldsymbol{\beta}$ fixed across the window and the residual white; it is pessimistic in seeing one window in isolation, whereas the smoother carries information across windows through the marginal prior of Eq. (28) and the coefficients are quasi-static far beyond one window. Its absolute level scales as $\sqrt{R}$, an assumed constant, and it measures one direction while DRMS is a two-axis RMS. What survives these caveats is the shape in W and the comparison across lines, and Section E uses it for exactly that: where the knee sits, and which line is weakest. The empirical ablation of Section F measures what the walk $\mathbf{Q}^{\beta}$, the inertial process factors, and the priors add.

Remark 1:
The condition is geometric: it depends on the map gradient under the flight path. Over a locally flat map, $\mathbf{G}\approx\mathbf{0}$, the projected gradient of Eq. (19) loses rank and position is unobservable regardless of the compensation: a map-information floor that bounds every estimator.

C. Factor graph formulation

The models of Section A induce four factor types on the variables $\boldsymbol{\chi}_{0:N}$, drawn in Fig. 3. Following the convention of Eq. (6), each is written as a squared Mahalanobis residual.

The *prior* factor anchors the initial joint state,

$$\bar{\phi}^{0}(\boldsymbol{\chi}_0) = \big\|\boldsymbol{\chi}_0 - \hat{\boldsymbol{\chi}}_0\big\|^2_{\mathbf{P}_0}, \tag{21}$$

where $\mathbf{P}_0$ is block diagonal in the inertial and compensation parts, with blocks $\mathbf{P}_0^{x}$ and $\mathbf{P}_0^{\beta}$.

The *inertial* factor relates consecutive states through Eq. (12),

$$\bar{\phi}_k^{\mathrm{ins}}(\boldsymbol{\chi}_{k-1}, \boldsymbol{\chi}_k) = \big\|\mathbf{x}_k - \boldsymbol{\Phi}_{k-1}\mathbf{x}_{k-1}\big\|^2_{\mathbf{Q}_{k-1}}. \tag{22}$$

The *compensation* factor imposes the random walk Eq. (15),

$$\bar{\phi}_k^{\beta}(\boldsymbol{\chi}_{k-1}, \boldsymbol{\chi}_k) = \big\|\boldsymbol{\beta}_k - \boldsymbol{\beta}_{k-1}\big\|^2_{\mathbf{Q}^{\beta}}, \tag{23}$$

so that the compensation is free to adapt along the flight at a rate set by $\mathbf{Q}^{\beta}$, and is not free to jump.

The *map* factor is the only one that couples the two chains,

$$\bar{\phi}_k^{\mathrm{map}}(\boldsymbol{\chi}_k) = \rho\Big(\big\|h(\bar{\mathbf{p}}_k+\delta\mathbf{p}_k)+\mathbf{A}_k^{\top}\boldsymbol{\beta}_k+S_k-z_k\big\|_R\Big), \tag{24}$$

with ρ a robust kernel discussed below. Collecting the four types gives the objective

$$\tilde{J}(\mathbf{X};\mathbf{Z}) = \bar{\phi}^{0}(\boldsymbol{\chi}_0)+\sum_{k=1}^{N}\bar{\phi}_k^{\mathrm{ins}}+\sum_{k=1}^{N}\bar{\phi}_k^{\beta}+\sum_{k=0}^{N}\bar{\phi}_k^{\mathrm{map}}, \tag{25}$$

minimized as in Eq. (7). The graph has two parallel chains, the inertial error and the compensation, tied together at every epoch by a map factor, and it is this tie that makes the estimation joint rather than sequential.

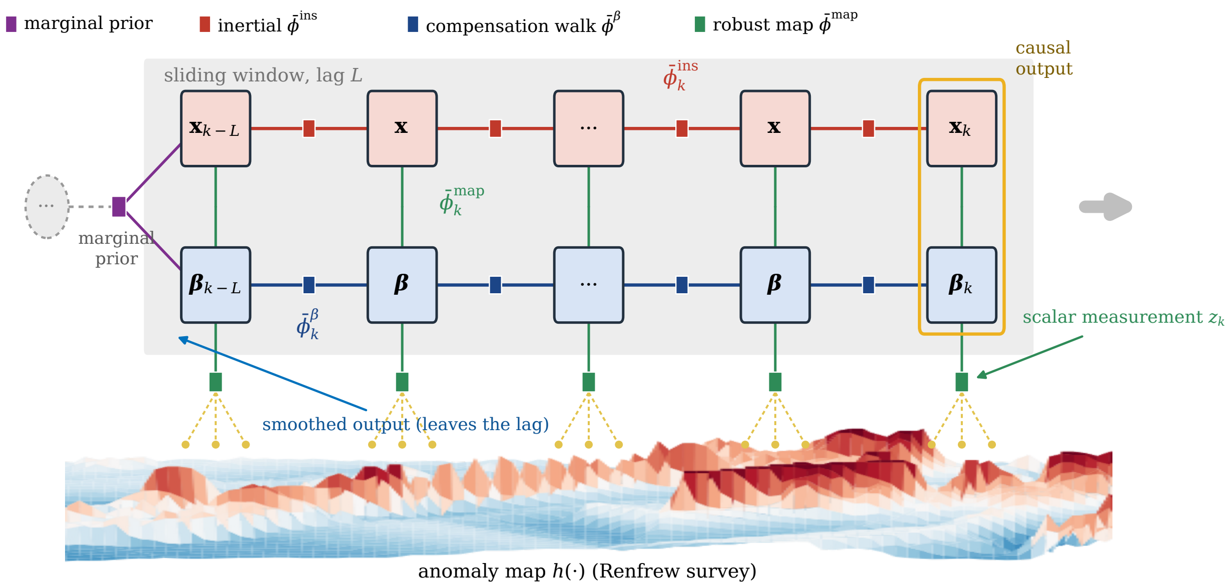

Fig. 3. Factor graph maintained by the smoother: two chains coupled at every epoch by the robust map factor carrying z_k; the marginal prior summarizes states older than the lag. Relief height is anomaly strength.

Map error and transient interference produce heavy-tailed residuals that a purely quadratic cost over-weights. We take ρ to be the Huber kernel [51]

$$\rho(u) = \begin{cases} \frac{1}{2}u^2, & |u| \le \delta, \\ \delta\big(|u| - \frac{1}{2}\delta\big), & |u| > \delta, \end{cases} \tag{26}$$

minimized by iteratively reweighted least squares, in which each iteration scales the map factor at epoch k by $w_k = \min(1, \delta/|u_k|)$ with $u_k = r_k/\sqrt{R}$, so that gross outliers are pulled toward an absolute-value penalty while inliers keep the efficient quadratic cost. We use $\delta = 1.345$, the standard choice for 95% Gaussian efficiency. Switchable constraints [52] and dynamic covariance scaling [53] fit the same factor interface and could replace Eq. (26) without touching the rest of the formulation.

D. Incremental fixed-lag solution

Minimizing Eq. (25) over the whole line is a batch problem, and by Section III it is available only after the flight. For onboard use we solve it at a fixed lag and incrementally.

1. *Fixed lag*

Let L be the lag, in seconds. At epoch k the graph retains the variables whose timestamps lie within L of the newest, and the variables older than that are marginalized: their information is compressed into a Gaussian factor on the oldest retained state, as described in Section III. Two estimates are therefore produced at every epoch, and they are different products.

- The *causal* estimate is the value of the newest variable χ_k immediately after the update. It uses no measurement taken after k, so it is available with zero delay and is the estimate that a real-time navigator would read.
- The *smoothed* estimate is the value a variable holds when it leaves the lag, after revision by every measurement in the following L seconds: the classical fixed-lag smoothed output. It is the more accurate of the two and it is available only L seconds late.

Both are components of one optimization. Writing J_k for the objective Eq. (25) restricted to the factors retained at epoch k, together with the marginal prior of Eq. (28) below, the smoother computes the joint estimate and reads two components of it,

$$\hat{\mathbf{X}}_k = \arg\min_{\mathbf{X}} J_k(\mathbf{X}), \quad \hat{\chi}_k^{\text{c}} = [\hat{\mathbf{X}}_k]_k, \quad \hat{\chi}_{k-\ell}^{\text{s}} = [\hat{\mathbf{X}}_k]_{k-\ell}, \tag{27}$$

with $\ell = L/\Delta$ the lag in states. Under a linear-Gaussian specialization these components coincide with the classical filtering and fixed-lag smoothing estimates, since a component of a Gaussian joint mode is the corresponding marginal mean; in the nonlinear robust formulation they are the current and delayed components of the fixed-lag joint MAP solution, and any covariance attached to them is the Laplace approximation at the current linearization and IRLS weights, without the M-estimator sandwich correction, hence if anything optimistic. Marginalization realizes the boundary compactly: if $\mathbf{x}_m$ denotes the states leaving the lag and $\mathbf{x}_b$ the retained states their factors touch, with joint information matrix and vector partitioned as $\boldsymbol{\Lambda} = \left[\begin{smallmatrix} \boldsymbol{\Lambda}_{mm} & \boldsymbol{\Lambda}_{mb} \\ \boldsymbol{\Lambda}_{bm} & \boldsymbol{\Lambda}_{bb} \end{smallmatrix}\right]$, $\boldsymbol{\eta} = \left[\begin{smallmatrix} \boldsymbol{\eta}_m \\ \boldsymbol{\eta}_b \end{smallmatrix}\right]$, then the marginal prior carried forward is the Schur complement

$$\bar{\boldsymbol{\Lambda}}_{bb} = \boldsymbol{\Lambda}_{bb} - \boldsymbol{\Lambda}_{bm}\boldsymbol{\Lambda}_{mm}^{-1}\boldsymbol{\Lambda}_{mb}, \qquad \bar{\boldsymbol{\eta}}_b = \boldsymbol{\eta}_b - \boldsymbol{\Lambda}_{bm}\boldsymbol{\Lambda}_{mm}^{-1}\boldsymbol{\eta}_m, \tag{28}$$

so for the linearized model no information is discarded. What is fixed is the linearization at which Eq. (28) is evaluated: for the nonlinear factors the carried prior is a

Laplace approximation, a potential source of overconfidence that the simulation of Section H finds below its resolution. This is the precise sense in which a smoothed state can no longer be revised. Equation (28) also requires $\mathbf{\Lambda}_{mm}$ to be invertible, which the process-noise floor of Section D guarantees.

Both are reported throughout Section V. Keeping them separate matters for comparison: a causal filter such as an EKF can only be compared against the causal estimate, whereas the smoothed estimate is the output of a system that is allowed to lag real time by L.

It is worth being precise about what the lag buys, because the two outputs depend on it very differently. Marginalization does not discard measurements, it compresses them, so the causal estimate at epoch k conditions on the same information, every measurement up to k, whatever the lag is. Lengthening the lag therefore should not improve it, and the only residual effect is that retained states may be relinearized more often. What the lag does change is how long a state stays revisable before Eq. (28) freezes it: the smoothed value for epoch $k-\ell$ conditions on measurements up to k, which the causal value for that epoch could not have used. The smoothed output is therefore not a delayed copy of the causal one but an estimate of a past epoch made with L seconds of hindsight, and the gap between the two is the price of requiring an answer immediately. Section E measures both dependences and finds them as described here.

2. Incremental update

Re-solving the retained interval from scratch at every epoch would repeat almost all of the previous work, since one new measurement disturbs only a small part of a long chain. We instead update the factorization in place using the Bayes-tree formulation of iSAM2 [36]. At each epoch the inertial factor Eq. (22), the compensation factor Eq. (23) and the map factor Eq. (24) of that epoch are added; iSAM2 identifies the variables whose linearization the new factors disturb, re-eliminates only that part of the tree, and marginalizes what has left the lag. The per-epoch cost is therefore set by the number of retained states, L divided by the state interval, and not by the flight duration.

3. State interval

Placing one variable per magnetometer sample makes the retained set large for any useful lag. We therefore place variables at an interval Δ longer than the sampling period, compounding the transition and accumulating the process noise exactly across the K samples that a variable interval spans,

$$\mathbf{\Phi}' = \mathbf{\Phi}_{k+K-1} \cdots \mathbf{\Phi}_k, \qquad \mathbf{Q}' \leftarrow \mathbf{\Phi}\mathbf{Q}'\mathbf{\Phi}^\top + \mathbf{Q}, \quad (29)$$

so that the retained chain is an exact marginal of the full-rate chain in its process model. The map factors at the intermediate samples are not carried. At survey speed the along-track anomaly signal lies well below the $0.5\,\mathrm{Hz}$ Nyquist frequency of the retained cadence, so the discarded samples are largely redundant, and denser sampling cannot substitute for a longer window: keeping all ten samples per state multiplies the rows of $\mathbf{G}$ and $\mathbf{\Psi}$ without changing the path they trace, so μ of Eq. (20) improves by exactly the $\sqrt{10}$ of independent averaging and its geometry not at all. That factor is not realized in flight, because at $67\,\mathrm{m\,s^{-1}}$ consecutive samples are under $7\,\mathrm{m}$ apart and share their map error; the ablation of Section F measures the outcome as a loss of $2.3\,\mathrm{m}$ on Mag 4 at four times the computation. Separability is bought with track, not with samples.

4. Conditioning

One numerical point matters for reproduction. The position channel of $\mathbf{Q}_k$ is minute, of order $10^{-31}\,\mathrm{rad}^2$ in the horizontal states, because position error is driven only through velocity over a single sample; whitening the inertial factor by $\mathbf{Q}_k^{-1/2}$ then spans some thirty orders of magnitude. A Cholesky factorization of the information matrix squares this conditioning and fails outright, returning an indeterminate variable. Two changes remove the failure: a floor of $10^{-20}\mathbf{I}$ added to $\mathbf{Q}_k$, which injects under a millimetre of position uncertainty per step, and a QR factorization of the square-root information matrix in place of Cholesky, which works on the square root and so tolerates the residual spread [34]. A larger floor, or a regularizing prior at every epoch, also suppresses the exception but pins the estimate onto the propagated INS and defeats the correction.

E. Prior selection

One element of Eq. (21) is set with the cold start in mind. The block $\mathbf{P}_0^\beta$ states what is believed about the compensation coefficients before any measurement, and at a cold start that belief must admit the field of an uncalibrated installation: the uncompensated cabin field is of order $2000\,\mathrm{nT}$, and the coefficients that explain it are far from zero in the units of Eq. (14). For initial stability we therefore widen the compensation block relative to a calibrated installation and use $\sigma_\beta = 100$ throughout. The isotropic form mixes the units of the coefficient groups; the dimensionally uniform alternative, a common prior contribution of $100\,\mathrm{nT}$ per regressor column, performs identically in the ablation. The setting is not delicate: the ablation of Section F varies it by an order of magnitude in each direction, and every other element of the configuration, and finds metre-level effects.

V. Experiments and Results

A. Dataset, maps, and protocol

All experiments use the public SGL 2020 challenge dataset [14]: a Cessna 208B instrumented with a flight-grade INS, several onboard scalar magnetometers (a compensated tail-stinger sensor and uncompensated cabin sensors Mag 4 and Mag 5, all optically pumped cesium-vapor units), a three-axis fluxgate, and reference truth. The flights serve distinct purposes, and we choose lines

TABLE I
Flight lines (FF free flight, SV survey; maps R Renfrew, E Eastern) and free-inertial drift.

Line	Purp.	Map	alt. [m]	dur. [min]	INS [m]
1003.02	SV	E	401	63.1	123.8
1003.08	SV	R	400	72.4	271.9
1007.02	FF	E	799	64.3	120.7
1007.06	FF	R	402	87.3	317.5

TABLE II
Proposed-estimator settings, one configuration for every case.

Group	Parameter	Value
Measurement	noise std. $\sqrt{R}$	$12\,\mathrm{nT}$
	robust kernel, Huber δ	1.345
FOGM S	(σ_S, τ)	$3\,\mathrm{nT}$, $180\,\mathrm{s}$
Initial $\mathbf{P}_0^x$	pos. / alt. std.	0.1 / $1.0\,\mathrm{m}$
	velocity std. (per axis)	$1.0\,\mathrm{m\,s^{-1}}$
Compensation	basis (App. B)	P+I+E+bias ($m = 19$)
	prior $\mathbf{P}_0^\beta = \mathbf{I}\sigma_\beta^2$	$\sigma_\beta = 100$
	walk $\mathbf{Q}^\beta$	$\mathbf{I}(1.0)^2\Delta t$
Smoother	lag L	$300\,\mathrm{s}$
	state interval Δ	$1\,\mathrm{s}$ ($K = 10$)
	solver	iSAM2, QR
	process-noise floor	$10^{-20}\mathbf{I}$
Evaluation	DRMS warm-up	$600\,\mathrm{s}$

accordingly (Table I). Flt1007 is the free-flight mission, in which the aircraft flies freely within the mapped areas, so its lines are the closest to a navigation task and are what prior work [14], [4], [19] reports; 1007.06 is our representative line for that reason. Flt1003 is a crustal-field survey flight, and its lines 1003.02 and 1003.08 are straighter than the free-flight legs. Flt1006 is a calibration-manoeuvre flight and, with the map-building flights (Flt1004–1005), is not used: its single $14\,\mathrm{min}$ line leaves too little to score after the warm-up. Two honest points remain. None of these lines is a dedicated long, straight-and-level operational transit; the free-flight and survey legs carry attitude variation, which stresses the compensation but also aids its observability by exciting the regressor, so performance on a pure operational cruise is not established here. And the lines span two altitude regimes, four near the $400\,\mathrm{m}$ survey-drape altitude and 1007.02 near $800\,\mathrm{m}$, where upward continuation attenuates the anomaly.

Map matching uses the Eastern and Renfrew high-resolution anomaly grids, upward-continued to flight altitude, with the IGRF core field restored so that measurement and map share an absolute scale. Cold start means the compensation coefficients are initialized at zero with no calibration flight; the navigation solution itself is accurately initialized (position standard deviation $0.1\,\mathrm{m}$, Table II), so the term denotes a compensation cold start, not an unknown-position start.

The error metric is the horizontal distance root-mean-square (DRMS) of position error, counted after a $600\,\mathrm{s}$ warm-up. Following Section D, the proposed estimator is reported twice in every table: the *causal* value, read at the newest state with no look-ahead, and the *smoothed* value, read as states leave the $300\,\mathrm{s}$ lag. The causal value is the like-for-like comparison against the recursive baselines, which also use no future data, and it is the number a navigator reads in flight; the smoothed value is the reconstructed trajectory, available L seconds after the fact, which is the quantity that matters wherever the answer is consumed after the epoch it describes: survey and map products, which are post-processed in any case, retrospective integrity checks on the causal solution, and any supervisory loop that can accept the latency. The look-ahead behind every reported number is stated in its table. One configuration (Table II) is used for every line and both magnetometers; nothing is tuned per case, and parameter sensitivity is quantified in the ablation of Section F. All numbers are produced by scripted runs under a fixed software environment and are available from the authors on request.

B. Baselines

Four references frame the comparison, all run by us on the same lines, the same maps, and the same inertial model. Table II describes the proposed estimator; the identically modelled online-TL EKF shares its compensation prior $\sigma_\beta = 100$, so that the margin between the two cannot be attributed to the prior, while the NN-augmented filter and the particle filter run the configurations stated below.

Free-inertial INS. The unaided INS drift (Table I) is the floor every aided estimator must beat, and the common reference that makes DRMS values comparable across lines of different length and roughness.

Online-TL EKF. The direct recursive counterpart of the proposed estimator: the same Pinson error state augmented with the same 19 time-varying TL coefficients, updated by the same scalar measurement, in an extended Kalman filter [15]. It shares every model term and the compensation prior with the factor graph, so the margin between the two isolates the estimator structure, recursive commitment against windowed re-linearization, from the modelling. The shared prior matters to the filter far more than to the smoother: at the reference implementation's calibrated-installation prior ($\sigma_\beta = 1$) the filter diverges on two Mag 4 lines and leaves the map on a third, the same prior sensitivity diagnosed for the particle filter below, and at $\sigma_\beta = 100$ it is bounded on all eight cases, the values reported in Table IV.

Online EKF+TL+NN. The strongest causal baseline we compare against augments the TL state with the weights of a small feed-forward network that absorbs what the linear basis cannot [15], [19]. We run the reference implementation of [15], tuned by us for a stable cold start, and do not claim to reproduce the extended filter of [19]. Cold start forces one substantive change: with no calibration flight to warm-start from, the network output normalization is initialized from the first five minutes of onboard interference; a naive port instead drives the network bias to $\sim 5 \times 10^4\,\mathrm{nT}$ and the filter to `NaN`. We keep the uncompensated scalar out of the network inputs: a position-dependent input carries the map gradient, so the network absorbs map signal and position drifts while the residual stays small (the scalar-augmented variant of [19] escapes this only by freezing the network after its transient). The tuned filter uses the permanent TL group with an eight-unit hidden layer. It is the one randomly initialized estimator in the comparison, so we quantify its seed spread on 1007.06: over ten seeds it reaches a median of $43.8\,\mathrm{m}$ (interquartile range $42.7\,\mathrm{m}$ to $45.5\,\mathrm{m}$) on Mag 4 and $18.2\,\mathrm{m}$ ($17.7\,\mathrm{m}$ to $18.5\,\mathrm{m}$) on Mag 5, consistent with published cold-start operating points [15], [19]; the tables quote one fixed-seed run per case ($48.8/18.3\,\mathrm{m}$ on this line).

Marginalized particle filter (MPF+TL). The nonparametric reference, included because particle methods are the standard resort when the map term is strongly nonlinear [8], [50]. Since the TL model is linear in its coefficients, $\boldsymbol{\beta}$ is conditionally linear-Gaussian, which motivates a Rao–Blackwellized implementation carrying the compensation and the remaining approximately linear states in conditional Kalman blocks: ours carries position in 1000 particles and marginalizes the remaining states, including $\boldsymbol{\beta}$, in the conditional Kalman block, with the fluxgate row entering the linear measurement Jacobian. Weights are computed in the log domain and position diversity is reinjected after resampling. The filter's cold-start behaviour is dominated by the compensation prior. Run at $\sigma_\beta = 1$, the scale of a calibrated installation, it fails at kilometre scale on every Mag 4 case: the conditional Kalman block then predicts a measurement variance orders of magnitude below the truth, and the particle weights degenerate into a directed walk up the local map gradient. Matching the prior to the installation scale ($\sigma_\beta = 700$, the magnitude the fitted coefficients reach) repairs the dominant term, bringing five of the eight cases to $25\,\mathrm{m}$ to $95\,\mathrm{m}$ (1227 to $93.5\,\mathrm{m}$ on 1007.06 Mag 4), but both sensors of 1007.02, the line with the weakest separability in Fig. 10, still diverge, and the results remain seed-sensitive. The contrast with the windowed estimator is the instructive part: a one-step filter must trust its compensation prior at every epoch, whereas the smoother amortizes the same prior over the window and is insensitive to it across two orders of magnitude (Section F). We therefore report the MPF in the text rather than in the comparison tables, and we do not claim it represents the best achievable particle filter. Attaching the neural network of the EKF+TL+NN to the particle filter does not rescue it and is sharply harmful on the well-conditioned sensor, degrading Mag 5 by one to two orders of magnitude across lines, the same direction seen for the NN-augmented EKF where the linear basis already suffices.

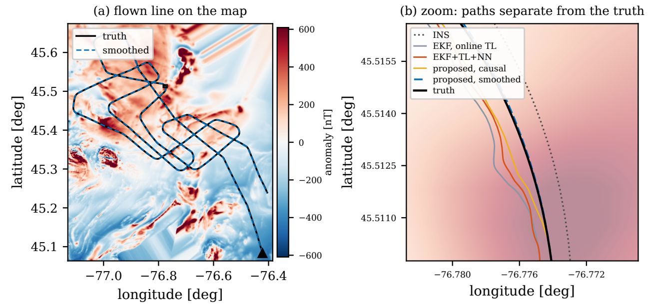

Fig. 4. Line 1007.06 Mag 4: (a) flown line with the zoom window; (b) the estimated paths separate from the truth, at the largest post-warm-up EKF error.

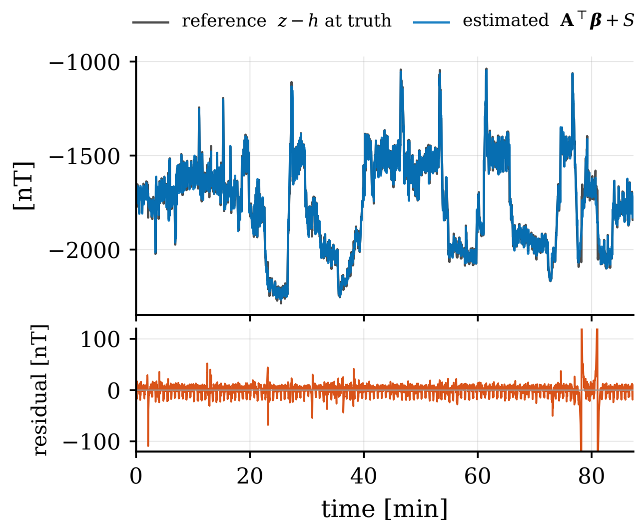

Fig. 5. Estimated $\mathbf{A}^\top\boldsymbol{\beta} + S$ against the interference reference, and the residual (line 1007.06 Mag 4).

C. Representative line

Table III reports the longest line, 1007.06 ($87\,\mathrm{min}$), on both cabin magnetometers, with the free-inertial drift at $317.5\,\mathrm{m}$. Read causally, with no look-ahead at all, the proposed estimator reaches $42.7\,\mathrm{m}$ on Mag 4 and $16.0\,\mathrm{m}$ on Mag 5, ahead of the online-TL EKF ($46.7/17.8\,\mathrm{m}$) and of the NN-augmented filter ($48.8/18.3\,\mathrm{m}$). The margin over the online-TL EKF is the purest number in this paper: the two estimators share the error model, the compensation model, the measurement model, and every noise setting, and differ only in that the filter commits each state once while the graph relinearizes and revises every state inside its lag. The margin over the EKF+TL+NN says something further: the network exists to absorb what the linear TL basis cannot, yet carrying the linear basis in a smoother beats carrying the linear basis plus a network in a filter, on both sensors, without any learned component. Allowed its $300\,\mathrm{s}$ lag, the smoothed estimate roughly halves the causal error, to $24.2/11.5\,\mathrm{m}$. Fig. 7 shows the error history, Fig. 6 its north and east components against every baseline,

TABLE III
Line 1007.06 cold start: DRMS [m] after a 600 s warm-up.

Estimator	Mag 4	Mag 5	look-ahead
EKF, online TL	46.3	17.6	none
EKF+TL+NN [15]	48.8	18.3	none
Proposed, causal	42.7	16.0	none
Proposed, 300 s **lag (smoothed)**	**24.2**	**11.5**	300 s

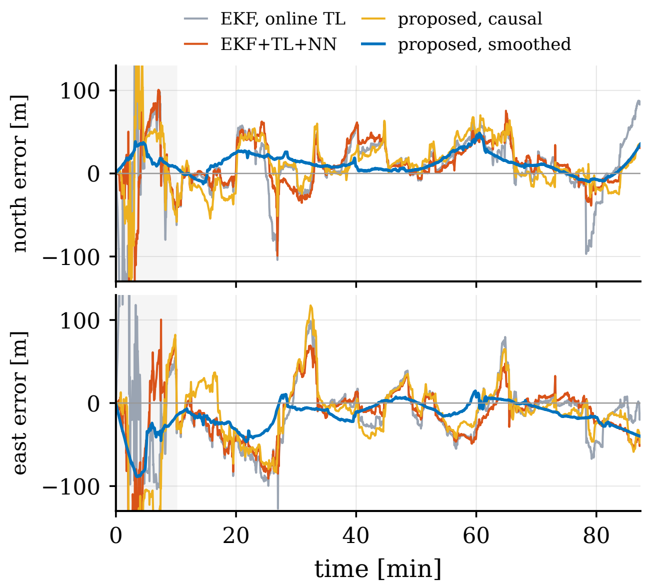

Fig. 6. North and east error histories, line 1007.06 Mag 4. The EKF+TL+NN is a fresh run (40.0 m here; ten-seed median 43.8 m).

Fig. 4 the recovered track on the map, and Fig. 5 the compensation the graph learned while recovering it.

D. Evaluation across lines and sensors

Table IV repeats the comparison on every counted line, spanning two flights, two maps, two altitude regimes, and both cabin magnetometers, with nothing re-tuned (Fig. 8). The smoothed estimate is the most accurate entry on all eight counted cases, between 10.5 and 28.4 m, its DRMS lower than that of the best causal baseline on the same case by factors of 1.5 to 5.2. Read causally, the proposed estimator is the best causal method on six of the eight; it trails the EKF+TL+NN on 1007.02 Mag 5 by 2.7 m and on 1003.08 Mag 4 by 2.3 m, and we report both rather than claiming a sweep. The margin is widest where the lines are hardest: on the two Mag 4 cases where the shared-prior EKF reaches 87.5 and 155.9 m, the smoothed estimate reads 28.4 and 19.6 m. The mechanism is the one Section A anticipated: the filter must explain the uncompensated field with whatever state is cheapest at each instant and commits immediately, while the smoother may revise its attribution for 300 s before committing; and the filter's dependence on the prior scale (Section B) is the recursive face of the same coin, since a one-step update

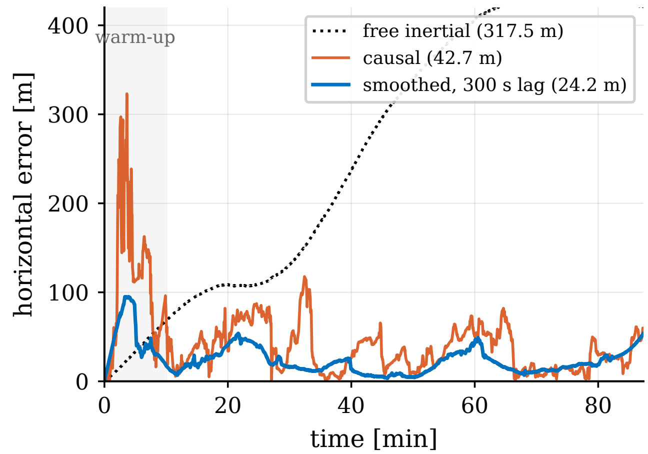

Fig. 7. Horizontal error, line 1007.06 Mag 4: the smoothed trace is the causal trace revised by the 300 s lag.

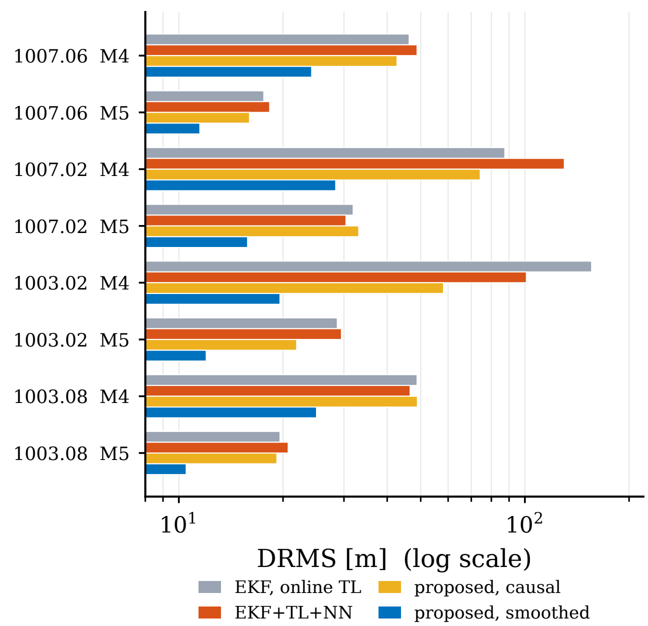

Fig. 8. Breadth comparison of Table IV, log axis.

has no window across which to amortize a mis-scaled prior.

E. Lag design

The lag is the estimator's principal design variable, and Table V sweeps it from 30 to 600 s on the four counted lines (Fig. 9). Two regularities carry the design guidance. First, the smoothed error improves monotonically with lag up to 300 s on every case, and weakly after it: lengthening the lag from 300 to 600 s, a doubling of both latency and computation, buys at most 2.6 m and on one case slightly degrades, while the step from 30 to 300 s buys between 3.6 and 41.8 m. The knee separates two regimes, and the separability metric of Eq. (20) says which is which: Fig. 10 evaluates $\mu^{-1/2}$ in sliding windows of length equal to

TABLE IV

Breadth: DRMS [m]; bold = best overall.

Line	Mag	EKF online	EKF+ TL+NN	Proposed causal	Proposed 300 s lag
1007.06	4	46.3	48.8	42.7	**24.2**
1007.06	5	17.6	18.3	16.0	**11.5**
1007.02	4	87.5	130.0	74.3	**28.4**
1007.02	5	31.9	30.4	33.1	**15.8**
1003.02	4	155.9	101.0	58.2	**19.6**
1003.02	5	28.7	29.5	21.9	**12.0**
1003.08	4	48.8	46.6	48.9	**25.0**
1003.08	5	19.6	20.7	19.2	**10.5**
best causal				**6 / 8**	
best overall					**8 / 8**

TABLE V

Lag sweep: smoothed DRMS [m]; the causal output varies by under a metre across the sweep.

Line	Mag	lag L [s], smoothed: 30	60	150	300	600	causal
1007.06	4	40.2	37.1	29.7	24.2	21.9	42.7
1007.06	5	15.1	14.3	12.6	11.5	11.4	16.0
1007.02	4	70.2	60.4	43.2	28.4	25.8	74.3
1007.02	5	27.9	24.6	18.3	15.8	14.2	33.1
1003.02	4	48.9	43.3	28.8	19.6	18.2	58.2
1003.02	5	20.1	18.3	14.9	12.0	10.6	21.9
1003.08	4	50.3	41.5	31.5	25.0	24.6	48.9
1003.08	5	17.4	15.8	12.5	10.5	10.7	19.2

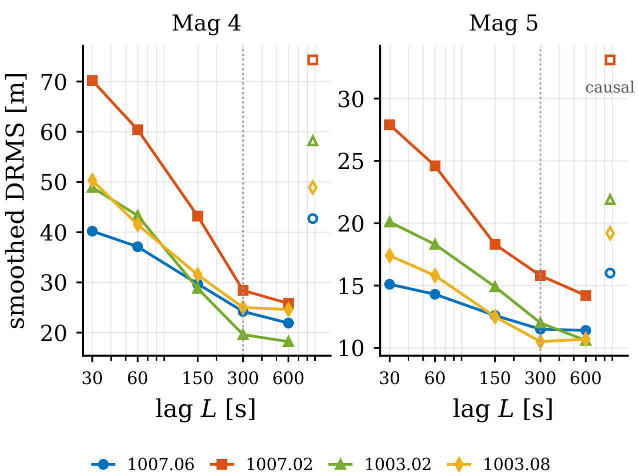

Fig. 9. Smoothed DRMS against lag; open markers show the lag-independent causal error.

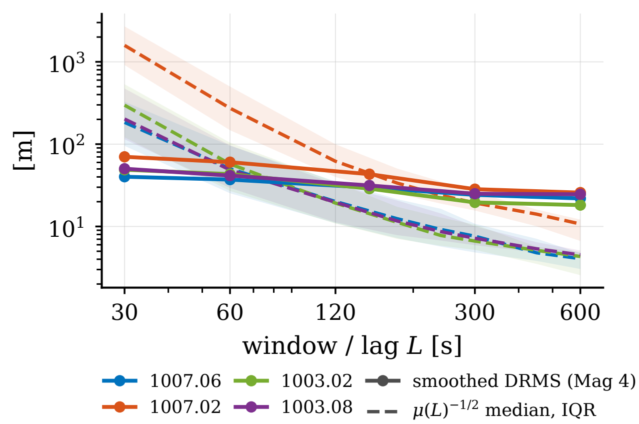

Fig. 10. Separability metric $\mu^{-1/2}$ of Eq. (20) at window length L (dashed, median and interquartile range over sliding windows) against the measured smoothed DRMS at lag L (solid, Mag 4; the metric is sensor-independent), per line.

the lag, at the $1\,\mathrm{Hz}$ kept cadence. Being a single-window quantity, the metric does not bound the smoothed error, which draws on information carried across windows by the marginal prior: at 30 s the per-line medians are $184\,\mathrm{m}$ to $1585\,\mathrm{m}$ while the measured smoothed DRMS is $15.1\,\mathrm{m}$ to $70.2\,\mathrm{m}$. It is also blind to the magnetometer, so the Mag 4 to Mag 5 gap is outside its reach. What it carries is the shape and the geography, and both match. The metric falls by one and a half to two orders of magnitude between 30 and 300 s, the range in which the measured smoothing gain is concentrated; past 300 s the measured DRMS stops improving even though the single-window metric keeps falling, so beyond the knee the error is no longer separability-limited but set by map error, inertial noise, and coefficient drift, which more look-ahead does not remove. And the metric identifies the weakest line: 1007.02, lowest at every window length (median $19.3\,\mathrm{m}$ at 300 s against $6.7\,\mathrm{m}$ to $7.6\,\mathrm{m}$ for the other three), is the line with the largest measured error at every lag on both sensors. The metric therefore explains the pre-knee regime and ranks the relative difficulty of the lines, but it is not by itself a predictor of the optimal lag, which is set by where the separability-limited regime ends. We take 300 s as the default. Second, the causal column is flat: across a twentyfold change in lag the causal error moves by less than a metre on every case. This is the behaviour Section D anticipates, and it is a check on the marginalization rather than a disappointment. Because the states leaving the lag are compressed into the marginal prior of Eq. (28) rather than dropped, the newest state conditions on the whole flight whatever the lag is, so only the retrospective revision has anything to gain from a longer window. The contrast with a sliding window that discards its history is instructive: in the GNSS/INS comparison of [54], where the graph retains only the states inside the window, the current-epoch error improves steadily as the window grows, from $5.18\,\mathrm{m}$ at $1\,\mathrm{s}$ to $3.65\,\mathrm{m}$ at the batch limit, because there the window length decides how much information the estimate has at all. Here it decides only how long a state is revised before it is frozen, so across this sweep the accuracy bought by look-ahead is smoothing gain to within a metre, and a designer who cannot afford latency loses essentially the gap between the two columns.

TABLE VI
Ablation on line 1007.06, one change per row: smoothed (causal) DRMS [m].

Element	Change	Mag 4	Mag 5
reference ($\sigma_\beta = 100$)		24.2 (42.7)	11.5 (16.0)
Comp. prior	$\sigma_\beta = 10$	24.1 (43.9)	11.5 (16.6)
	$\sigma_\beta = 1000$	23.9 (42.1)	11.2 (15.4)
	per column, $100\,\mathrm{nT}$	23.7 (42.7)	11.6 (16.1)
Comp. walk	$\mathbf{Q}^\beta \times 0.01$	26.3 (56.4)	12.7 (20.0)
	$\mathbf{Q}^\beta \times 100$	24.1 (42.7)	12.4 (25.8)
FOGM	$\sigma_S \times 10$	23.9 (38.8)	11.0 (16.6)
Robust kernel	removed	25.4 (42.9)	11.7 (16.5)
	Huber $\delta = 10$	25.6 (42.8)	11.7 (16.5)
Meas. noise	$\sqrt{R} = 65\,\mathrm{nT}$	24.6 (42.9)	12.7 (23.7)
Process floor	$10^{-16}\mathbf{I}$	33.1 (74.7)	13.1 (19.3)
Relin.	every variable, every epoch	24.1 (42.6)	11.5 (16.0)
Meas. density	all 10 samples per state	26.5 (44.2)	11.7 (16.0)

F. Ablation

Table VI varies each element of the configuration on line 1007.06, one at a time. The estimator is insensitive at the metre level to the modelling parameters, while excessive numerical regularization through the process-noise floor produces a larger degradation. The compensation prior may move an order of magnitude in either direction, or be replaced by a per-column prior that assigns each regressor column $100\,\mathrm{nT}$ of measurement contribution, for changes under a metre. Slowing the compensation walk a hundredfold costs about two metres, consistent with a real installation state that drifts in flight; speeding it up costs little. Removing the Huber kernel or widening its threshold costs one to two metres, so robustness to map error helps but does not carry the result. Inflating the assumed measurement noise fivefold, or relinearizing every variable at every epoch, changes the estimate by less than half a metre; the latter confirms that iSAM2's selective relinearization gives up nothing here. The largest single effect is the process-noise floor: raising it from 10^{-16} to 10^{-12} times the nominal position channel, i.e. from $10^{-20}\mathbf{I}$ to $10^{-16}\mathbf{I}$, costs $8.9\,\mathrm{m}$ on Mag 4, which is why Section D treats the floor as a conditioning device to be kept minimal rather than a tuning knob. Attaching all ten raw measurements to each state instead of one costs $2.3\,\mathrm{m}$ on Mag 4 and nothing on Mag 5, at four times the computation, supporting the $1\,\mathrm{Hz}$ state cadence.

G. Cold-start transient and initial uncertainty

Every DRMS so far excludes the first $600\,\mathrm{s}$, so Table VII reports what happens inside that window, when the compensation is least known. The smoothed value of an epoch is the value it holds when it leaves the lag, so every entry in the smoothed columns has been revised by up to $300\,\mathrm{s}$ of later measurements. The estimator does not pass through a divergent phase: the smoothed output runs at $10.7\,\mathrm{m}$ to $55.0\,\mathrm{m}$ DRMS during the transient, within $30\,\%$ of its own converged value on six of the eight cases and within a factor of 2.3 on the remaining two. What the transient taxes is causality. The causal output peaks between 57 and $323\,\mathrm{m}$, and over the same interval the free-inertial drift is only $33\,\mathrm{m}$ to $65\,\mathrm{m}$ DRMS with peaks of $64\,\mathrm{m}$ to $116\,\mathrm{m}$: while the window is too short for position and compensation to be separable (Proposition 1 at small W), the causal estimate can attribute interference to position in a way the unaided INS cannot, and on most cases it is transiently worse than coasting. The misattribution is repaid within the lag, since the smoothed revision of the same epochs is already at the converged scale, and after the warm-up the causal output beats the free-inertial drift on every counted case.

TABLE VII
First $600\,\mathrm{s}$ of the cold start: DRMS and peak horizontal error [m] inside the warm-up window.

		smoothed		causal	
Line	Mag	DRMS	peak	DRMS	peak
1007.06	4	55.0	95.0	135.5	323.1
1007.06	5	10.7	18.6	41.2	107.6
1007.02	4	36.7	73.3	93.2	220.3
1007.02	5	36.5	53.2	65.2	134.4
1003.02	4	21.5	48.9	72.3	206.8
1003.02	5	11.9	15.2	40.0	107.6
1003.08	4	23.3	31.4	110.6	303.6
1003.08	5	13.2	18.8	31.0	57.1

The cold start of this paper assumes a known takeoff position ($0.1\,\mathrm{m}$ horizontal prior) and an unknown installation field. Relaxing the former tests how much the results lean on it: widening the initial horizontal position prior to 10, 50, and $100\,\mathrm{m}$ on both sensors of 1007.06 moves the post-warm-up smoothed DRMS by at most $0.4\,\mathrm{m}$ (24.19 to $24.54\,\mathrm{m}$ on Mag 4, 11.53 to $11.66\,\mathrm{m}$ on Mag 5) and the causal DRMS by at most $4.6\,\mathrm{m}$. The transient pays more: at the $100\,\mathrm{m}$ prior the in-window smoothed DRMS rises from 55.0 to $69.1\,\mathrm{m}$ on Mag 4 and from 10.7 to $22.0\,\mathrm{m}$ on Mag 5, while the causal transient is essentially unchanged. The two cases of Table VII that beat their converged values do so by leaning on the tight anchor, and the advantage disappears with it. The information that converges the estimator is the map gradient accumulated over the first window, not the decimetre anchor, and the post-warm-up claims do not depend on a surveyed takeoff point.

H. Statistical consistency

Whether the reported covariance can be believed is a separate question from accuracy, and we answer it in two parts, formal and diagnostic.

Monte Carlo under a perfect model. The error-state truth is simulated from the estimator's own discretized model on the 1007.06 geometry, the transition and process noise of Eq. (29) at the $1\,\mathrm{Hz}$ cadence, and measurements are synthesized from the same map, regressor, and R with Gaussian noise; the robust kernel is disabled, since the noise is truly Gaussian, and the smoother runs exactly as flown otherwise. For each epoch the two-degree-of-freedom horizontal-position NEES $\mathbf{e}^\top \mathbf{P}^{-1}\mathbf{e}$ is evaluated with the marginal covariance $\mathbf{P}$ from the Bayes tree, at the newest state for the causal output and at the state leaving the lag for the smoothed output. The synthetic measurements are generated through the nonlinear map at the perturbed position, so linearization error is present in the experiment. Over 40 independent runs the causal average NEES is 2.10 ± 0.13 and the smoothed average 2.11 ± 0.18, mean and 95% t-interval over run means, against the consistent value of 2; within-run samples are serially correlated over the lag, so the run-level spread, not a per-sample chi-square band, is the honest error bar. Both intervals contain 2: the solver machinery, including the marginalization of Eq. (28), is statistically consistent when the model is true, and any overconfidence from the frozen linearizations is below the resolution of the test, under about ten percent in the NEES (variance) sense. The test exercises the Gaussian specialization; the robust kernel of the flown configuration is outside its scope.

Coverage in flight. On real data the model is not perfect, the map carries error and the interference is not white, so the same statistic is a diagnostic of the reported covariance rather than a consistency test. On 1007.06, the fraction of post-warm-up epochs whose horizontal error lies inside the 95% ellipse of the reported Laplace covariance (χ^2_2 threshold 5.99) is $93\,\%$ causal and $62\,\%$ smoothed on Mag 5, and $53\,\%$ causal and $27\,\%$ smoothed on Mag 4; the errors are correlated over roughly the lag, so these fractions carry an effective sample size of order tens, not thousands. The pattern is informative: the covariance is nearly honest where the model is nearly right, and optimistic exactly where unmodeled map error and residual interference dominate, more so for the smoothed output whose reported uncertainty is smaller. Restoring $95\,\%$ coverage would take a standard-deviation inflation of roughly 1.1 (Mag 5 causal) to 3 (Mag 4 smoothed); an integrity layer built on this estimator should inflate accordingly, and the accuracy claims of this paper rest on the measured DRMS, not on the reported covariance.

I. Computational cost

All timings are single-core, on a laptop-class x86 processor under WSL2, with the measurement factor evaluated through a Python callback; they are upper bounds on a native implementation. At the reference configuration the incremental update takes $7.3\,\mathrm{ms}$ to $9.0\,\mathrm{ms}$ per state across the eight counted line–magnetometer cases, mean $8.2\,\mathrm{ms}$, so the $87\,\mathrm{min}$ line completes in $42\,\mathrm{s}$ and the margin to the $1\,\mathrm{Hz}$ state cadence exceeds two orders of magnitude. The cost scales with the number of retained states, so halving the lag halves it, and the two expensive variants of the ablation bound the alternatives: relinearizing every variable at every epoch costs $68\,\mathrm{ms}$ per state and attaching all ten raw measurements $31\,\mathrm{ms}$, both still real time at $1\,\mathrm{Hz}$ but paying roughly ninefold and fourfold for accuracy changes the ablation shows to be, respectively, below half a metre and a two-metre degradation.

VI. Conclusion

This paper posed cold-start magnetic-anomaly navigation as a single estimation problem: an aircraft with an uncalibrated magnetometer installation must recover its position from a scalar reading in which the map signal and the aircraft's own field are mixed, with no calibration flight and no satellite aiding. The formulation carries the Tolles–Lawson compensation coefficients as time-varying variables of a factor graph alongside the inertial error state, ties the two chains through the map measurement, and solves the graph with an incremental fixed-lag smoother built on iSAM2, at a $300\,\mathrm{s}$ lag and a $1\,\mathrm{Hz}$ state cadence. The estimator produces two outputs, a causal estimate with no look-ahead and a smoothed estimate revised by the full lag, and we report both throughout, because they answer different questions: the causal output is what a navigator reads in flight, the smoothed output is what a system that may lag real time delivers.

On the SGL 2020 flight data the answers are as follows. Against a recursive EKF carrying the identical models, the causal output alone is more accurate on six of eight cold-start cases, so windowed relinearization pays before any look-ahead is spent. Against an NN-augmented online EKF baseline, an EKF augmented with an online neural network, the linear-basis smoother is more accurate on both magnetometers of the representative line with no learned component at all. The smoothed output is the most accurate estimate on all eight counted cases, between 10.5 and $28.4\,\mathrm{m}$ where free-inertial drift is hundreds of metres, reducing DRMS by factors of 1.5 to 5.2 relative to the best causal baseline. The lag sweep locates the design knee near $300\,\mathrm{s}$, about $20\,\mathrm{km}$ of track. In the tested cases, lengthening the lag acts almost purely as a purchase of smoothing accuracy with latency, since the causal DRMS varies by less than a metre across the sweep. The ablation finds no delicate modelling parameter, apart from the numerical process-noise floor that must be kept minimal: one configuration serves every line and both sensors, and the update costs $7.3\,\mathrm{ms}$ to $9.0\,\mathrm{ms}$ per state in an unoptimized implementation, two orders of magnitude inside real time.

Several limitations bound the study. Each line is a single physical realization, so every DRMS is a point estimate; the statistical statements are counts across cases, not distributions. The reported covariance is consistent under matched-model Monte Carlo simulation, but the flight coverage diagnostic shows substantial overconfidence where unmodeled map error and residual interference dominate, most of all for the smoothed Mag 4 output, so the covariance should not be used as an integrity bound as it stands. The baselines are our own implementations under one shared protocol; the particle-filter reference in particular is reported as unstable under per-case tuning, and we do not claim it represents the best achievable particle filter. None of the lines is a long, straight-and-level operational transit, so performance under the weak regressor excitation of a pure cruise is not established. And the compensation prior that a cold start needs is a measured setting rather than a derived one: we show a broad insensitive range, but give no rule for choosing its scale from installation data.

These limitations set the agenda: robust covariance calibration and integrity monitoring under model mismatch, a transit-like flight regime, sparser instantiation of the compensation chain in the manner of [50], and a native-factor implementation that would carry the real-time margin from the state cadence down to the raw sensor rate. A further direction composes naturally with this work rather than competing with it: the inertial chain could be carried on the group of extended poses in the manner of [27], whose benefit is that the propagation Jacobians stop depending on the estimate. That is attractive here for a reason specific to a smoother, since it is at the marginalization boundary, where Eq. (28) freezes a linearization, that estimate-dependent Jacobians do the most damage, and the flight coverage of Section H is where such a change would show up.

Appendix A
Pinson Error-State Dynamics

The navigation-error block of $\mathbf{\Phi}_t$ in Eq. (12) follows the Pinson model in a local north-east-down frame [2], [1]. With position, velocity, and attitude errors $(\delta\mathbf{p}, \delta\mathbf{v}, \boldsymbol{\psi})$ and sensor biases $(\mathbf{b}_a, \mathbf{b}_g)$, the continuous error dynamics are

$$\begin{aligned} \dot{\delta\mathbf{p}} &= \mathbf{F}_{pp}\,\delta\mathbf{p} + \delta\mathbf{v}, \\ \dot{\delta\mathbf{v}} &= \mathbf{F}_{vp}\,\delta\mathbf{p} + \mathbf{F}_{vv}\,\delta\mathbf{v} - [\mathbf{f}^n\times]\,\boldsymbol{\psi} + \mathbf{C}_b^n\mathbf{b}_a, \\ \dot{\boldsymbol{\psi}} &= \mathbf{F}_{\psi p}\,\delta\mathbf{p} + \mathbf{F}_{\psi v}\,\delta\mathbf{v} - [\boldsymbol{\omega}_{in}^n\times]\,\boldsymbol{\psi} - \mathbf{C}_b^n\mathbf{b}_g, \end{aligned} \tag{30}$$

where $\mathbf{f}^n$ is the specific force, $\mathbf{C}_b^n$ the body-to-nav rotation, $\boldsymbol{\omega}_{in}^n$ the inertial rate, $[\cdot\times]$ the skew operator, and the $\mathbf{F}_{\bullet\bullet}$ blocks collect the transport-rate, gravity-gradient, and Coriolis terms. The biases follow first-order Gauss–Markov or random-walk models, as does the scalar magnetic disturbance state S with correlation time τ. The discrete $\mathbf{\Phi}_t = \exp(\mathbf{F}_t\,\Delta t)$ and $\mathbf{Q}_t$ are obtained by standard van Loan discretization [48]. Only $\mathbf{g}_t^\top$, the position row of the linearized measurement Eq. (17), couples this block to the map.

Appendix B
Tolles–Lawson Basis

The basis row $\mathbf{A}_t$ of Eq. (14) is built from the fluxgate direction cosines $\hat{\mathbf{b}} = [\,u\ v\ w\,]^\top$ (with $u^2+v^2+w^2=1$) and the scalar field magnitude B [11], [12]. The permanent moment contributes the three cosines (u, v, w); the induced magnetization, symmetric in the field, contributes the six products $B\,(u^2, uv, uw, v^2, vw, w^2)$; and the eddy-current field, proportional to the rate of change of the cosines, contributes the nine products $B\,(u\dot{u}, u\dot{v}, \ldots, w\dot{w})$. Stacking these gives an eighteen-term basis, optionally augmented by a constant bias, and $c_t = \mathbf{A}_t^\top\boldsymbol{\beta}_t$ is linear in the coefficients $\boldsymbol{\beta}_t$, the property that lets them enter the factor graph as ordinary linear-measurement variables. In the cold-start regime the calibration flight that classically fixes $\boldsymbol{\beta}$ is unavailable, which is precisely why we carry it as a graph variable.

Appendix C
Proof of Proposition 1

Over the window, a position perturbation $\delta\mathbf{p}$ and a coefficient perturbation $\delta\boldsymbol{\beta}$ change the stacked residual by $\mathbf{G}\,\delta\mathbf{p} + \mathbf{\Psi}\,\delta\boldsymbol{\beta}$. The perturbation $\delta\mathbf{p}$ is indistinguishable from a compensation adjustment exactly when some $\delta\boldsymbol{\beta}$ cancels it, that is when $\mathbf{G}\,\delta\mathbf{p} = -\mathbf{\Psi}\,\delta\boldsymbol{\beta} \in \mathrm{range}(\mathbf{\Psi})$, which proves the first statement. For the second, decompose $\mathbf{G}\,\delta\mathbf{p} = \mathbf{\Psi}\mathbf{\Psi}^\dagger\mathbf{G}\,\delta\mathbf{p} + (\mathbf{I} - \mathbf{\Psi}\mathbf{\Psi}^\dagger)\mathbf{G}\,\delta\mathbf{p}$: the first term always lies in $\mathrm{range}(\mathbf{\Psi})$, so $\mathbf{G}\,\delta\mathbf{p} \in \mathrm{range}(\mathbf{\Psi})$ if and only if $(\mathbf{I} - \mathbf{\Psi}\mathbf{\Psi}^\dagger)\mathbf{G}\,\delta\mathbf{p} = \mathbf{0}$. Hence every nonzero $\delta\mathbf{p}$ is distinguishable if and only if the projected matrix $(\mathbf{I} - \mathbf{\Psi}\mathbf{\Psi}^\dagger)\mathbf{G}$ has trivial kernel, that is rank two. ■

Acknowledgment

The experiments use the publicly released SGL 2020 challenge dataset [14].

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